

Objective

To implement and demonstrate the `NullPointerException` in Java using try-catch and handle the exception without terminating the program.

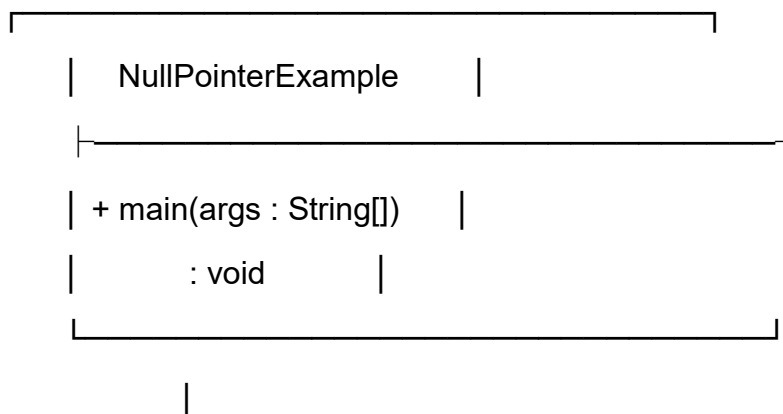
Steps for Execution

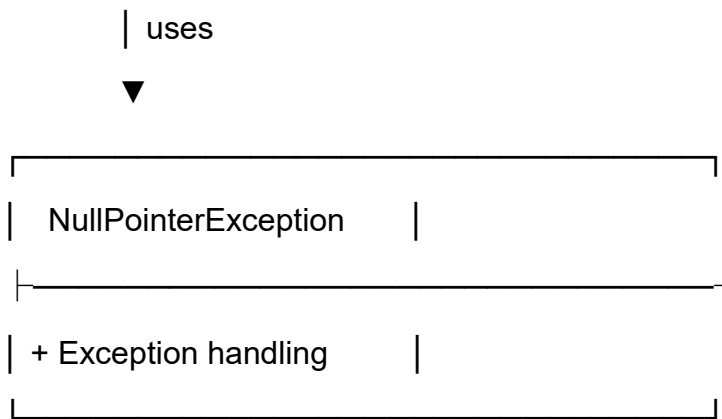
1. Create a Java project and package named `javacore`.
2. Create a file named `NullPointerExceptionExample.java` and enter the given code.
3. Declare a `String` variable with null value.
4. Use try-catch to handle the exception.
5. Compile and run the program.
6. Observe the output.

Algorithm

1. Start the program.
2. Declare name as null.
3. Try to find the length of name.
4. If `NullPointerException` occurs, catch it.
5. Display the error message.
6. Display "Program ended".
7. Stop the program.

UML CLASS DIAGRAM:





Relationship

NullPointerExceptionExample contains the main() method.

The program handles NullPointerException using a try-catch block.

The exception occurs because name contains a null value and length() is called on it.

CODE:

```
package javacore;

public class NullPointerExceptionExample {

    public static void main(String[] args) {

        String name = null;

        try {
            System.out.println("Name: " + name);
            System.out.println("Length: " + name.length());
        }
        catch (NullPointerException e) {
            System.out.println("Error: Name is null.");
        }

        System.out.println("Program ended.");
    }
}
```

```
}  
}
```

OUTPUT:

Name: null

Error: Name is null.

Program ended.

Result

The program successfully demonstrates `NullPointerException` handling in Java. Since `name` is null, calling `name.length()` causes a `NullPointerException`, which is handled using the catch block, allowing the program to continue normally.